

Part 1: Security Features, Mechanisms, and Protocols

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
WhatsApp	Confidentiality	End-to-End Encryption	Signal Protocol	Keeps messages private between the sender and receiver.
Netflix	Authorization	Access Tokens	JWT	Checks whether the user has permission to watch the content.
Bitcoin	Integrity	Hashing	SHA-256	Makes it difficult to change past transactions.
GitHub	Authentication	Key Pair	SSH	Allows secure login using a private and public key.
Cloudflare	Availability	Traffic Distribution	Anycast Routing	Sends traffic to different servers to keep websites running.

Part 2: Security Controls, CIA, Mechanism, Prevention, Detection, Recovery

Application	Security Control	CIA	Mechanism / Protocol	Prevention	Detection	Recovery
WhatsApp	End-to-End Encryption	Confidentiality	Signal Protocol	Prevents others from reading messages.	Security alerts help identify suspicious activity.	Encrypted backups help restore chats.
Netflix	Access Control	Confidentiality	JWT, HTTPS/TLS	Prevents unauthorized account access.	Detects unusual login activity.	Account recovery helps regain access.
Bitcoin	Hashing	Integrity	SHA-256	Prevents unauthorized changes to transactions.	Blockchain records can be checked for changes.	Copies of the blockchain maintain the correct record.
GitHub	SSH Keys & Access Control	Confidentiality	SSH, HTTPS	Prevents unauthorized access to repositories.	Activity logs help identify suspicious access.	Account recovery and repository backups help restore data.
Cloudflare	DDoS Protection & Traffic Distribution	Availability	Anycast, CDN	Protects websites from traffic-based attacks.	Monitors unusual traffic patterns.	Redirects traffic to available servers.